

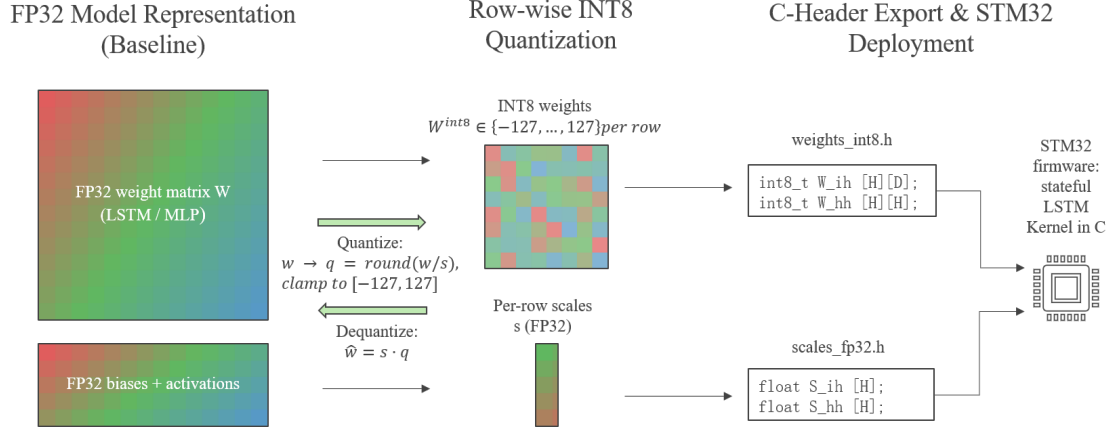


Figure 7.5: Row-wise weight-only INT8 quantization and embedded export used for the recurrent matrices. Floating-point matrix rows are represented by integer codes and individual floating-point scales. Biases and activations remain in floating point, while the stored recurrent weight matrices require fewer bytes.

The complete mixed-precision gate pre-activation can be written as

$$\mathbf{a}_t = \mathbf{s}_{ih} \odot (W_{ih}^{int8} \mathbf{x}_t) + \mathbf{s}_{hh} \odot (W_{hh}^{int8} \mathbf{h}_{t-1}) + \mathbf{b}_{fp32}. \quad (7.17)$$

In the inspected C kernels, each INT8 recurrent weight is converted to FP32 and multiplied by its row scale inside the accumulation. Only W_{ih} and W_{hh} use INT8 storage. The scales, biases, hidden and cell states, gate activations, MLP weights and arithmetic, and final output remain FP32. The recurrent matrices account for about 80% of the raw Base constants, so this boundary targets the dominant persistent storage while avoiding an additional quantization source in the recurrent state. It is therefore a weight-only mixed-precision implementation, not a full-integer network, and it cannot deliver the RAM and arithmetic benefits of activation and state quantization. Figure A.11 makes this precision boundary explicit.

The exported parameters are written as C headers and integrated into manually implemented stateful LSTM + MLP firmware. This route exposes all scale operations and state updates and avoids ambiguity in generated support for continuous recurrent state.[133, 134, 135] Streaming checks against the source implementation typically produce MAE differences below 10^{-4} .